

MODULE 5: SOFTWARE VERIFICATION AND VALIDATION
WEEK 5

Learning Outcomes:

After completing this course, you are expected to demonstrate the following:

1. Tackle specific verification and validation requirements for manufacturing and quality systems.

A. Engage

Trivia



Figure 5.0

System software is software designed to provide a platform for other software. Examples of system software include operating systems like macOS, Linux, Android and Microsoft Windows, computational science software, game engines, industrial automation, and software as a service application.

B. Explore

Video Titles:

1. Difference between verification and validation in software testing in Software Engineering.
2. JavaScript Form Validation.

YouTube Links:

1. <https://www.youtube.com/watch?v=ujx322POCQU>
2. <https://www.youtube.com/watch?v=ln0nB0ABaUk>

C. Explain

Software Engineering | Verification and Validation

Verification and Validation is the process of investigating that a software system satisfies specifications and standards and it fulfills the required purpose. Barry Boehm described verification and validation as the following:

Verification: Are we building the product right?

Validation: Are we building the right product?

Verification:

Verification is the process of checking that a software achieves its goal without any bugs. It is the process to ensure whether the product that is developed is right or not. It verifies whether the developed product fulfills the requirements that we have. Verification is Static Testing.

Activities involved in verification:

- ✓ Inspections
- ✓ Reviews
- ✓ Walkthroughs
- ✓ Desk-checking

Validation:

Validation is the process of checking whether the software product is up to the mark or in other words product has high level requirements. It is the process of checking the validation of product i.e. it checks what we are developing is the right product. it is validation of actual and expected product. Validation is the Dynamic Testing.

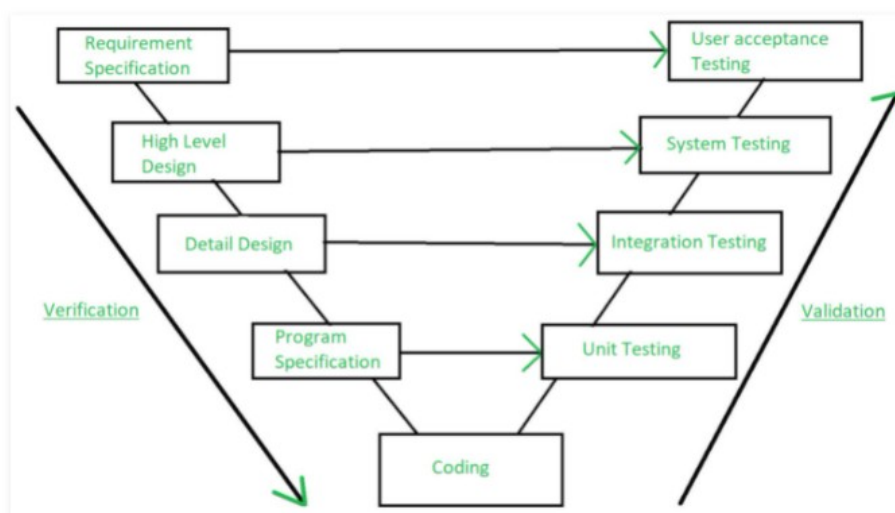


Figure 5.1

Activities involved in validation:

- ✓ Black box testing
- ✓ White box testing
- ✓ Unit testing
- ✓ Integration testing

D. Elaborate

Verification and Validation Testing

In this section, we will learn about verification and validation testing and their major differences.

Verification testing

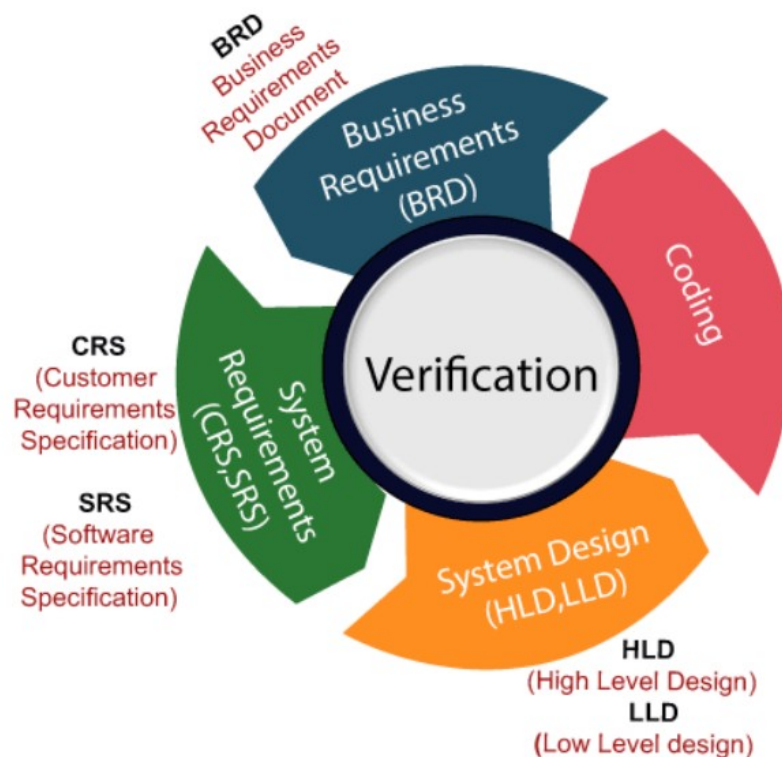


Figure 5.2

Verification testing includes different activities such as business requirements, system requirements, design review, and code walk-through while developing a product. It is also known as static testing, where we are ensuring that "we are developing the right product or not". And it also checks that the developed application fulfilling all the requirements given by the client.

Validation testing

Validation testing is testing where tester performed functional and non-functional testing. Here functional testing includes Unit Testing (UT), Integration Testing (IT) and System Testing (ST), and non-functional testing includes User acceptance testing (UAT).

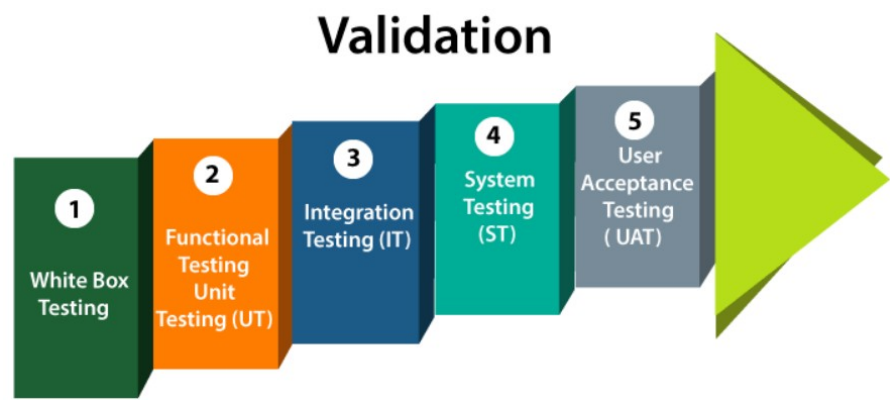


Figure 5.3

Validation testing is also known as dynamic testing, where we are ensuring that "we have developed the product right." And it also checks that the software meets the business needs of the client.

Difference between verification and validation testing

Verification	Validation
We check whether we are developing the right product or not.	We check whether the developed product is right.
Verification is also known as static testing .	Validation is also known as dynamic testing .
Verification includes different methods like Inspections, Reviews, and Walkthroughs.	Validation includes testing like functional testing , system testing, integration , and User acceptance testing.
It is a process of checking the work-products (not the final product) of a development cycle to decide whether the product meets the specified requirements.	It is a process of checking the software during or at the end of the development cycle to decide whether the software follow the specified business requirements.
Quality assurance comes under verification testing.	Quality control comes under validation testing.
The execution of code does not happen in the verification testing.	In validation testing, the execution of code happens.
In verification testing, we can find the bugs early in the development phase of the product.	In the validation testing, we can find those bugs, which are not caught in the verification process.

Verification testing is executed by the Quality assurance team to make sure that the product is developed according to customers' requirements.	Validation testing is executed by the testing team to test the application.
Verification is done before the validation testing.	After verification testing, validation testing takes place.
In this type of testing, we can verify that the inputs follow the outputs or not.	In this type of testing, we can validate that the user accepts the product or not.

E. Evaluate

ASSESSMENT:

Instructions: Write your answer on the Answer Sheet (AS) provided in this module. (50 pts)

1. Create a table that shows the differences between verification and validation testing

LABORATORY ACTIVITY 5.0

1. Give a sample code using HTML and JavaScript which uses an input validation in a form, your resources can be found at video resources folder, Week5_Software_Verification_and_Validation name of the video is “JavaScript Form Validation”. Good luck.

References:

- 1. *Software Engineering | Verification and Validation*
- Link
- <https://www.geeksforgeeks.org/software-engineering-verification-and-validation/>
- 2. *Verification and Validation Testing*
- Link
- <https://www.javatpoint.com/verification-and-validation-testing>

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